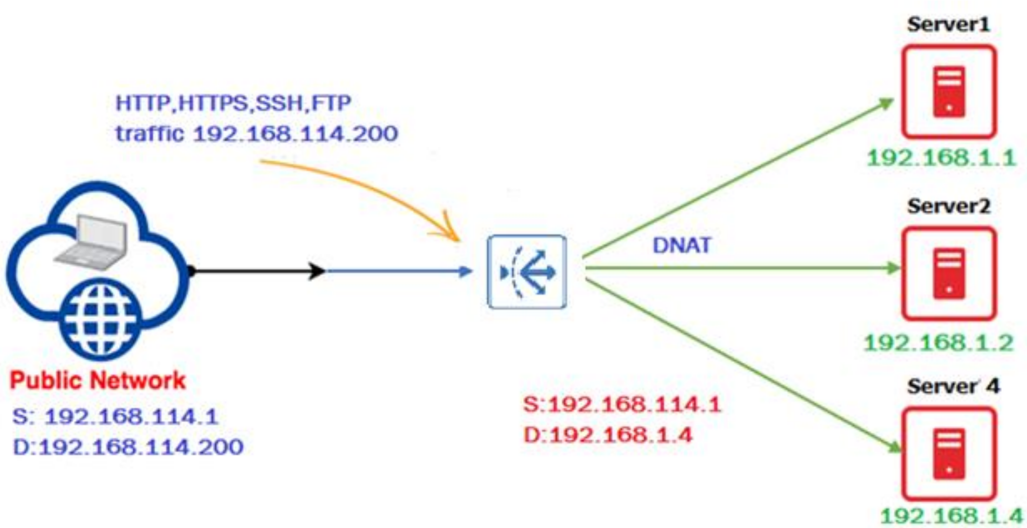
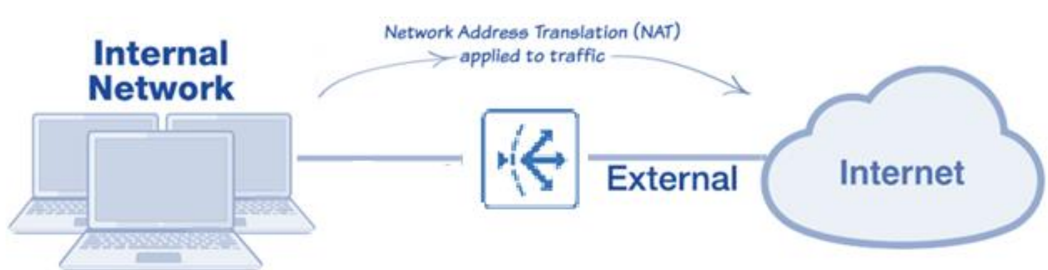
Network Address Translation:

- **NAT** stands for **Network Address Translation**.
- It is a process that changes **private IP addresses** to **public IP addresses**.
- NAT allows a single device to act as a bridge between a private network and internet.
- NAT helps save public IP addresses and enables many devices on a local network (LAN) to use a single public IP to access the internet.
- It's mostly used on **routers** and **firewalls** to perform IP address translations.
- NAT improves security by hiding the internal network's structure and private addresses.
- It helps connect private devices to the internet and ensures that the internal network isn't directly visible from the outside.
- **Source NAT (SNAT)** and **Destination NAT (DNAT)** are two main types of NAT
- These configurations control how the source & destination IP addresses are translated.
- NAT is essential in managing IP address use and enhancing security in networks.





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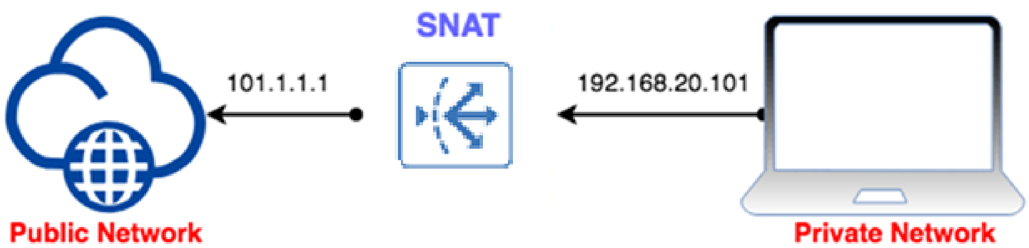
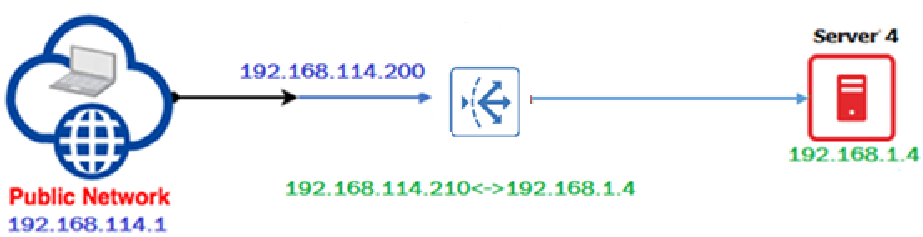
Source Network Address Translation (SNAT):

- SNAT Source NAT, is abbreviation for Source Network Address Translation.
- Translates the packet header source IP address to the configured address.
- Source NAT changes source address in Internet Protocol header of a packet.
- Source NAT may also change the source port in the TCP and UDP headers.
- It is used when an internal host needs to initiate connection to an external host.
- It is used when a private host needs to initiate a connection to a public host.
- Device performing NAT changes private IP address of source host to public IP address.



1-to-1 NAT:

- **One-to-One OR 1-to-1 NAT** maps a **public IP address** to a **private IP address**.
- Maps the public IP address for an interface to an IP address on a private network.
- It allows internal servers with private IPs to be accessible from the internet using a public IP.
- External traffic sent to the public IP is forwarded to the mapped internal private IP.
- Responses from the internal server are sent back through the public IP.
- Exposes only the mapped public IP to external clients, protecting the internal network.
- Simplifies the exposure of internal services such as web server to the internet.
- In FortiADC 1-to-1 NAT requires you to specify both the public and private IP addresses.
- One-to-One NAT in FortiADC contains rules for static bidirectional NAT translation.



Port Forwarding:

- In Port Forwarding maps an external published protocol port to the actual port.
- Acts as a bridge between the external internet and internal network services.
- Port Forwarding modifies both the Layer 3 and 4 header of OSI Reference model.
- It allows specific UDP or TCP port on global address to be translated to specific port.
- Configuration for port forwarding is included in the configuration for 1-to-1 NAT.